The Little Hands of Antwerp
Preliminary Palaeographic Description of the Tentative Hands R and S

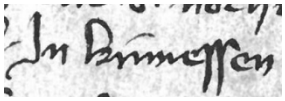
This appendix has been contributed by: Jan Burgers, Godfried Croenen, Els De Paermentier and Robin Waeytens

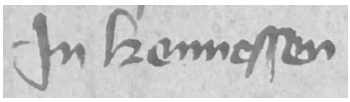
In the main text of the paper, tentative identifications of two larger scribal oeuvres (R and S) are proposed. In this appendix, we offer more detailed paleographic descriptions of these handwritings. We will argue, for each grouping, why the identified variants, despite internal differences, may nevertheless be attributed to a single scribal hand. This approach is based on the principle that individual handwriting features are not necessarily static but may display a degree of variation within the output of a single scribe. Such variation may be explained by, among other factors, the chronological development of the handwriting, adaptation to changing writing conditions, differences in writing materials or writing speed, the degree of formality of the document as a reflection of the importance of the recorded transaction or the parties involved, or the influence of contemporary scribal models. When the overall appearance of the script and its principal morphological characteristics remain chronologically consistent, and any deviations are confined to a limited number of letter forms or minor graphical details, it is reasonable to consider all such variants as the work of a same hand. In other words, the attributions proposed here are based on the combined occurrence of multiple corresponding features. The presence of both stable and more variable script forms within the same chronological context supports the hypothesis that these differences do not necessarily indicate different hands, but may instead represent normal variation that can be expected within the output of a single scribe.

R comprises a large and diverse subset of charters originating from 10 of the scribal clusters originally proposed by the HDBSCAN algorithm applied to the features extract by the custom model (the unclustered documents [HDBSCAN label −1], H5, H7, and H12–18); the earliest attestation of which dates to 17 March 1311 (#17_03_1311-01 (H7)) and the latest to 24 August 1348 (#24_08_1348-01 (H14)). Some variants occur only within a relatively brief period of one to four years (H12, 1314–1317; H5, 1323; H17, 1334–1337), whereas others span several decades (H7, 1311–1347; the unclustered documents, 1311–1345). Although the script forms in these documents display a high degree of similarity according to our analyses, they also exhibit a number of significant differences that make it possible to trace the development of this hand over time. There are several reasons why the possibility should not be excluded that these documents are attributable to a single hand, which may plausibly be identified as that of the aldermen's clerk Jan van Boendale:

- many of the variant groups overlap chronologically, strengthening the possibility that a single hand may evolve over time while nevertheless retaining the same idiosyncratic features across several decades;
- the palaeographical characteristics of the various clusters display striking similarities to the linguistic features that the computational analysis assigned to R, regardless of the palaeographical variant to which they belong;
- the written forms of several recurrent formulaic expressions (*dictamen*) in the aldermen's deeds exhibit strong morphological similarities irrespective of the palaeographical variant cluster to which they were assigned by the computer. This is particularly evident in phrases such as *scepenen in Antwerpen* and

maken cond in the intitulatio and notification formulae, and *in kinnessen van desen letteren* and *ghegheven int jaer* in the corroboratio formula. Interestingly, it would seem this scribe seminally introduced the corroboratio with the spelling *in kinnessen* (see #10_04_1319-01 (H15), dated 1319, and #09_12_1328-01 (H7), dated 1328), but from 1331 onwards he consistently adopted the spelling *in kennessen* (see #17_02_1331-01 (H16), dated 1331). This shift suggests an orthographic development within an otherwise stable scribal practice rather than a change of hand;





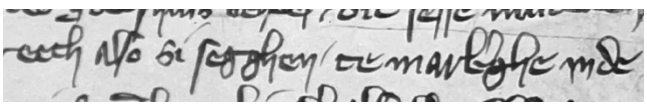
- the striking overlap between the period during which this hand cluster was active and the lifetime and documented career of Jan van Boendale as an aldermen's clerk further supports this attribution.

Cluster R: Temporal range of the palaeographical variants		
Scribal variant	**earliest attestation**	**latest attestation**
unclustered	1311	1345
H5	1323	1323
H7	1311	1347
H12	1314	1317
H13	1321	1338
H14	1338	1348
H15	1319	1328
H16	1326	1333
H17	1334	1337
H18	1334	1338

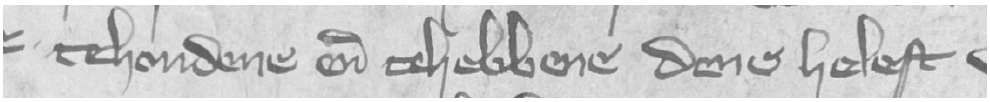
R – Palaeographic description

The script of hand R is characterised by:

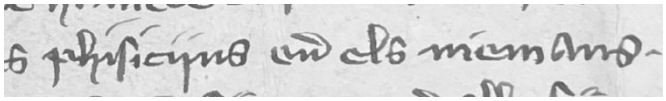
- a regular, upright (diplomatic) minuscule script, which, when executed more rapidly, tends to slant to the right (e.g. #28_11_1338-01 (unclustered), dated 1338);



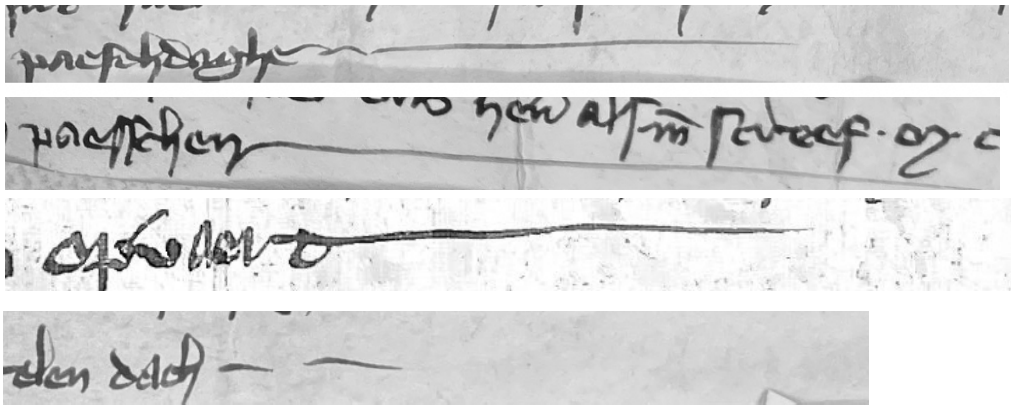
- a pronounced contrast between bold strokes and hairlines (e.g. #09_03_1320-01 (H15) dd. 1320);



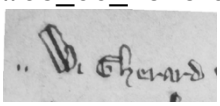
- the frequent, though not systematic, use of diacritical marks, such as fine hairline tittles placed above the letter *i* (e.g. #04_11_1334-01 (unclustered) dd. 1334);



- the recurrent use of line fillers in the form of a slightly undulating horizontal stroke on the document-final line, which, if the letter form permits, is often executed as a continuation of the final stroke of the last letter of the preceding word (e.g. #08_04_1328-02 (unclustered) dd. 1328; #28_04_1337-01 (H18) dd. 1337; #17_05_1343-01 (H14) dd. 1343; #14_07_1336-01 (H18) dd. 1336);

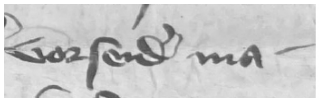


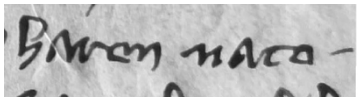
- the systematic positioning of a decorative mark in the upper-left corner of the text (e.g. #20_06_1332-01 (H16) dd. 1332; #09_09_1314-01 (H12); #05_06_1315-01 (H12));



- a set of morphologically distinctive letter forms, which will be discussed in greater detail below;

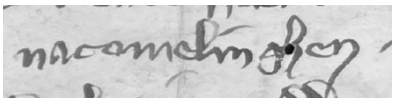
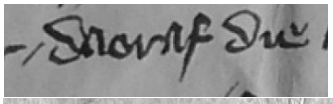
- from the 1320s onwards, a more systematic use of hyphens for words split over the line can be observed (bv. #08_04_1328-02 (unclustered) dd. 1328; #07_02_1333-01 (H16) dd. 1333; #16_07_1335-01 (H17) dd. 1335);

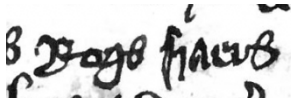






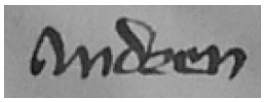
- from the 1330s onwards, the hand regularly can exhibit a slight tremor, particularly in the ascenders. This feature may be due to a neurological condition, although it could equally have resulted from a less stable writing surface or otherwise suboptimal writing conditions (e.g. #07_02_1333-01 (H16) dd 1333; #16_07_1335-01 (H17) dd 1335; #12_06_1346-02 (H7) dd 1346 (‘*daeraf*’); #07_06_1348-01 (H14) dd 1348).

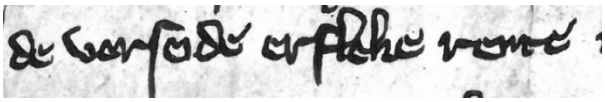


Characteristic morphological features of hand R:

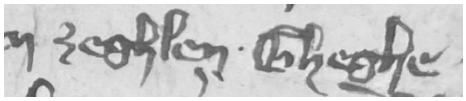
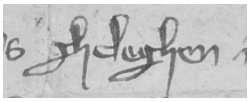
- a cursive *a*, whose lobe usually remains open at the bottom and which frequently displays a small (inside or outside) loop or an upward stroke on the left-hand side (e.g. #11_08_1332-01 (H7) dd. 1332; #07_02_1333-01 (H16) dd. 1333);

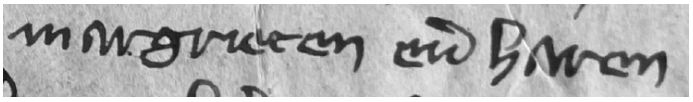
- an *e* whose lobe remains open and which — particularly when occurring as the final letter of a word — ends in a strongly pronounced upward final straight stroke or tongue (e.g. #04_03_1338-01 (unclustered) dd. 1338);



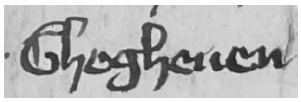
- a *g* with a short, rounded, and usually closed lower bow, alternating with a variant in which the tail forms an angular turn below the baseline and becomes somewhat longer (e.g. #71_05_03_1339-01; #79_05_10_1339-01; #156_10_06_1341-01). The elongated lower bow of *g*, extending horizontally along the writing line, is found predominantly in the earliest phase, up to c. 1320, after which the smaller and more rounded form becomes the dominant variant (see respectively #19_01_1317-01 (H12) dd. 1317-01 and #12_07_1334-01 (H17) dd. 1334);

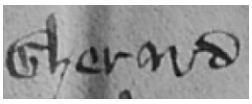



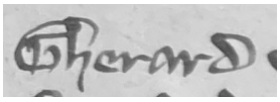
- a v-shaped *r* in which the final stroke is detached from the stem and is formed by an ascending pen stroke rising from the baseline (e.g. #16_07_1335-01 (H17) dd. 1335);

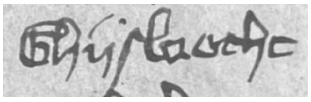


- a majuscule *G* whose left downstroke is doubled with a decorative stroke and which features, at its upper end, a left-curving hairline, so that the pen needs to be lifted for formation of the following letter (e.g. '*Ghegeven*' (*passim*); '*Gherard*' in #20_06_1323-01 (H5) dd 1323 and #17_02_1331-01 (H16) dd 1331; '*Ghijsbrecht*' #09_07_1336-01 (H18) dd 1336);



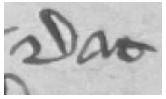






- a majuscule variant of *d* starting with a 2-shaped introductory stroke, which remains open at the top, then curves upwards to the left from the baseline before descending in a bold downstroke (e.g. #29_03_1321-01 (H7) dd. 1321; #09_12_1328-01 (H7) dd. 1328; #15_04_1330-01 (H16) dd. 1330);

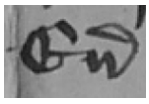






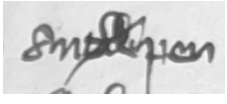
- a rounded majuscule *E* whose down stroke is doubled and which has a prominently extended tongue (e.g. #11_08_1332-01 (H7) dd. 1332; #29_02_1326-01 (H13) dd. 1326; #01_08_1336-01 (H18) dd. 1336 ('*elleve*');







- characteristic abbreviations such as the diagonal stroke through the *w* in *Antwerpen* on the first line from approximately 1335 onwards (e.g. #28_11_1338-01 (unclustered) dd. 1338).



S – Palaeographic description

Hand S comprises a grouping of deeds from primarily four HDBSCAN clusters (H4, H6, H8, and H9). A substantial proportion of the documents assigned to these groups exhibit the same orthographic and dialectal features and, on palaeographical grounds, may likely be attributable to a single scribal hand. The earliest document in the cluster dates from November 1327 (H9), while the most recent dates from June 1355 (H4). From a palaeographical perspective, variants H6 and H9 display the greatest morphological similarity to one another and together cover virtually the entire attested chronological range (1327–1354). H9, however, is primarily represented by the earliest documents, dating from the late 1320s and the 1330s. This chronological distribution may suggest either a gradual development of a single hand over time or the persistence of a stable scribal practice across several decades.

Cluster S: Temporal range of the palaeographical variants		
Scribal variant	**earliest attestation**	**latest attestation**
H4	1348	1355
H6	1341	1354
H8	1345	1355
H9	1327	1346

The script represented by the scribal variants assigned to Cluster S is characterised by a degree of variability in several letter forms, which appears in part to result from differences in the level of cursiveness, understood here as the extent of connections formed within and between letters without lifting the pen. Some of the documents belonging to this cluster are written in a highly cursive script, characterised by a greater number of hairlines and extended ending strokes, whereas others are more posed and executed in a more deliberate manner, resulting in a comparatively austere and stable appearance.

The more cursive documents are found in H4 and in approximately half of the documents assigned to H9, while the more formal script predominates in H6 and H8. In the more formal manifestation of the script, the scribe shows a preference for the single-compartment *a*, whereas in the more cursive documents the double-compartment *a* occurs more frequently. The forms of *g* and *v* likewise appear to be closely related to the degree of the script's cursiveness exhibited. Importantly, these variable letter forms are almost invariably used alongside one another within individual documents, even if particular deeds may display a preference for one variant over another. A chronological evaluation of both the stable and the variable features suggests that this hand continued to develop new variants until the 1330s, after which its graphical repertoire gradually became more stable and less variable.

This hand may be characterised by a number of both stable and variable features, which are discussed below (most examples are taken from one of the earliest surviving charters, #01_10_1354-02 (H4) dd. 1328):

- a highly variable degree of slant, ranging from an almost upright script to forms slanting markedly to the right;



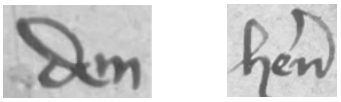
- a *k* with a high-set lobe and a short horizontal ending stroke;



- a long, thin, and slightly ascending macron;



- the digraph *en*, alternating between a ligatured cursive form and a consciously disjoined executio;



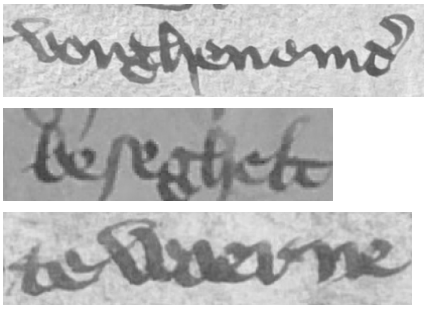
- the letter combinations *ee* and *ec*, which are never connected cursively, the upper sections of the letters remaining separate while their lower parts touch;



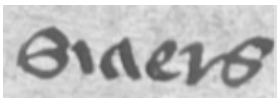
- the letter *a,* which appears in both a single-compartment variant with an inclined stem and a double-compartment variant, the latter occurring in both cursive and more posed executions;



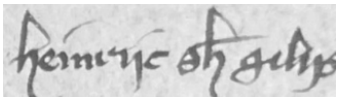
- the letters *h* (#06_02_1346-01 (H9) dd. 1346)*, l* (#20_04_1349-01 (H6) dd. 1349) and *w* (#06_03_1349-01 (H8) dd. 1349), which are cursively connected to the preceding letter;



- the round *s* at the end of a word, which slants to the right and exhibits a closed loop; in contrast, the round *s* at the beginning of a word is typically broader and more upright (see the word *siaers*); the distinction between initial and final *s*, however, becomes somewhat less clear in the final period of Hand S (see, for example, line 5 of document #23_07_1349-01 (H6));

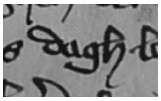


- single diacritical marks above the letter combination *ii* and *ij* (e.g. #30_11_1327-01 (H9); but not in #29_06_1328-01 (H9));

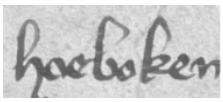


- the limb of *h,* which either extends straight below the baseline or curves gently to the right in a downwards ending stroke (e.g. #21_07_1334-01 (H8) dd. 1334; #19_07_1341-01 (H6) dd. 1341);

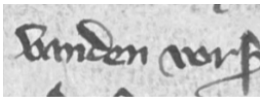




- the cursive loops of *b, h, k* and *l*, which may take either an triangular, sail-shaped form or a more rounded shape;



- the letters *v* and *w,* whose first stroke begins at the baseline with either a small curved introductory stroke or an upward loop (e.g. #30_11_1327-01 (H9); the *v* without loop does not occur in in #29_06_1328-01 (H9));

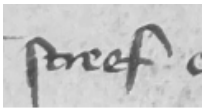


- the letter *p*, with either a curved, pointed descender or, less commonly, a shorter and straighter descender connected cursively to the second stroke of the letter;

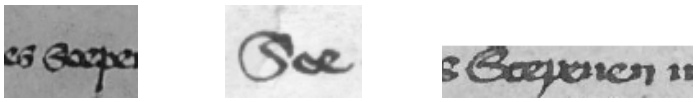




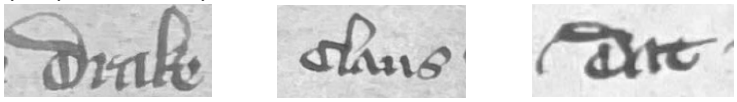
- the contrasting slanting directions of the bold strokes of *s* and *f* in the word *screef* in the dating clause, where the *s* is almost consistently inclined to the left and the *f* to the right, together with the characteristically bold horizontal bar of the latter;



- the characteristic round 6-form *s* at the beginning of a word, formed with a pronounced right-hand curve that folds back in a sharp angle towards the left (#27_01_1341-01 (H6) dd. 1341; #03_05_1341-01 (H9) dd. 1341; #01_08_1353-01 (H6) dd 1353);



- the doubling of the bows and lobes with a decorative downstroke of rounded (majuscule) letters such as *D*, *C* and *O* (cf. Willem Drake, #19_11_1334-02 (H9) dd. 1334; 'Claus', #05_09_1336-01 (H9) dd. 1336; 'dat', #20_05_1336-01 (H9) dd. 1336);

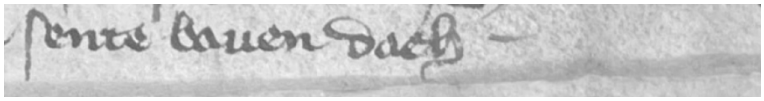


- the numbers M CCC in the dating clause, which are consistently written as Roman numerals, while the remainder of the year is spelled out in words (e.g. #19_11_1334-02 (H9) dd. 1334);



- the termination of the text without punctuation, or, in some cases, marked by a single dot or a short horizontal stroke (e.g. #30_05_1352-01 (H6) dd. 1352; #02_10_1347-01 (H8) dd. 1347).





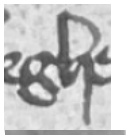
Over the nearly thirty years during which this scribe can be traced, the script remains remarkably consistent, including its repertoire of variant letter forms. During the 1330s, however, the upper part of the descender of the tall *s* generally becomes somewhat shorter than before, often extending only marginally above minim height. Around 1350, the ascenders of *b, h, k*, and *l* likewise tend to become shorter. At the same time, the characteristic *k* — whose lobe remains positioned high above the baseline — frequently develops a more diagonally descending end-stroke, which occasionally touches the foot of the stem, itself often prolonged upwards. Furthermore, the spelling *ij* marked by a single stroke as diacritic gradually disappears and is commonly replaced by a double *ii* with two separate tittles, although these are typically absent in the opening word *Wij*. These developments, however, represent only minor changes within an otherwise highly stable scribal practice and do not fundamentally alter the overall appearance of the hand.

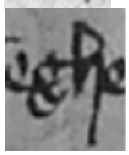
A clear development can be observed in the letter *g*.

- 1a) In the earliest documents (H9), the *g* is executed in two strokes, with the second stroke frequently extending through the lower loop to connect it to the following letter, especially *h* (example from #29_06_1328-01 (H9));



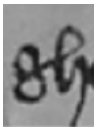
- 2a) In the 1330s, the loop of the tail of *g* becomes flatter and more strongly oriented towards the right. The ductus of the letter grows more complex, and the *g* is typically executed in four or five pen strokes. The cursive connection with the following letter no longer proceeds through the tail loop but instead originates from a transitional stroke extending from the upper part of the upper lobe (example from #05_09_1336-01 (H9));





- 2b) In the 1340s, the same basic form remains in use, but the right-hand side of the tail loop is now often finished with a slight upward return stroke (example from #07_05_1343-01 (H6));

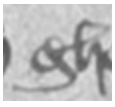
- 1b) At the same time, the earliest form (1a) reappears, although it is now executed with a more pronounced rightward orientation than in the 1320s (example from #07_05_1343-01 (H6));



- 1c) The same form also occurs with a more widely extended tail loop drawn towards the right, now exhibiting sharp angular turns on both the left and right sides. In this variant, the upper bow of the *g* is occasionally left open (example from #25_04_1346-01 (H6));



- 3) In the 1340s, a further variant of *g* appears that is structurally related to type 1. The bow and tail are formed in a cursive stroke with a small loop at the upper right of the upper lobe. Below the baseline, the tail extends into a broad,



flattened horizontal stroke, while the loop remains open. As in variant 1b, the cursive link with the following letter, frequently *h*, proceeds from a transitional stroke issuing from the upper part of the bow (example from #25_04_1346-01 (H6)).

By the 1350s, the scribe appears to have developed a preference for variants 1a and 1c, although the other forms continue to occur as well.

H4 distinguishes itself by a somewhat more cursive script than that found in the other clusters, characterised by the frequent use of long ending strokes — sometimes extending horizontally — attached to the tails of *h*, *m* and *n* at the end of a word, and *y* (e.g. #17_06_1354-01 (H4), dated 1354). In addition, this variant makes frequent use of a double-compartment *a*, often executed without cursive ligatures (e.g. #01_10_1354-02 (H4), dated 1354), and of a ending-*e* provided with an upward-curving serif (e.g. #09_07_1349-01 (H3), dated 1355). Because of this increased cursiveness and its somewhat 'baroque' hairlines, the script creates an impression that differs from that of the other clusters. Nevertheless, all of its letter forms correspond to variants attested elsewhere within Cluster S. Compare, for example, charter #05_02_1351-01 (H4), which exhibits the characteristic *k*, the various forms of *g*, both the upright and slanting forms of *s* in *siaers*, the long and slender abbreviation strokes, the variable degree of slant, and other features found throughout the cluster.





